

ECON 1550 International Finance

# Output and the Exchange Rate in the Short Run

# Behavioral equations in the goods market

$$Y = C + I + G + CA$$

$$\begin{array}{l} \text{DEMAND FOR} \\ \text{DOMESTIC} \\ \text{GOOD} \end{array} = C + I + G + CA$$

$$\begin{array}{l} \text{SUPPLY FOR} \\ \text{DOMESTIC} \\ \text{GOOD} \end{array} = Y$$

# Demand for domestic goods

$$\begin{array}{l} \text{DEMAND} \\ \text{FOR DOM} \\ \text{GOODS} \end{array} = C + I + G + CA = D$$

$$\begin{array}{l} \text{DOMESTIC} \\ \text{DEMAND} \\ \text{FOR GOODS} \end{array} \equiv A = C + I + G$$

$$A + EX - IM = A + CA = D$$

# Demand for domestic goods

$$C = C(Y_D^{(+1)}) \quad \text{CONSUMPTION (OF DOM. AND FOREIGN)}$$

$$Y_D \equiv Y - T \quad \text{DISPOSABLE INCOME}$$

$$I = \bar{I} \quad \text{EXOGENOUS INVESTMENT}$$

$$G = \bar{G}$$

"

GOV. PURCHASES

} BOTH DOM  
AND FOR.  
GOODS

$$CA = CA(q, Y - T, Y^*)$$

$$q \equiv \frac{EP^*}{P} = \frac{\text{DOLLAR PRICE OF FOREIGN GOOD}}{\text{DOLLAR PRICE OF DOM GOOD}}$$

# Demand for domestic goods

$$CA = EX - IM$$

$$EX = EX(q, Y^*)$$

(+1)   (+1)

$$IM = IM(q, Y_D)$$

(-)   (+1)

$q \uparrow$  VOLUME & PRICE  $q \uparrow$   
**VOLUME DOMINATES**

$q \uparrow$  REAL DEPR.

$$\frac{EP^*}{P} = q$$

$\Rightarrow$  HOME  
 GOOD  
 APPEARS  
 CHEAPER

$$\underbrace{IM}_{\text{DOM GOOD}} = \underbrace{\frac{\text{PRICE}}{\text{DOM GOOD}}}_{\text{FOREIGN GOOD}} \times \underbrace{VOLUME}_{\text{FOREIGN GOOD}}$$

# A short-run model of the goods market

| Exogenous variables |                       |
|---------------------|-----------------------|
| Variable            | Description           |
| $E$                 | Nominal exchange rate |
| $I$                 | Investment            |
| $G$                 | Government spending   |
| $T$                 | Taxes                 |
| $P$                 | Price level           |
| $P^*$               | Foreign price level   |
| $Y^*$               | Foreign income        |

## Endogenous variables

| Variable | Description               | Equation                                                                                                                                                                        | Type of equation      |
|----------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| $Y$      | Income, production        | $Y = D$                                                                                                                                                                         | Equilibrium condition |
| $Y_D$    | Disposable income         | $Y_D \equiv Y - T$                                                                                                                                                              | Identity              |
| EX       | Exports                   | $EX = EX(q, Y^*)$<br><span style="margin-left: 150px;">(+)</span> <span style="margin-left: 100px;">(+)</span>                                                                  | Behavioral            |
| IM       | Imports                   | $IM = IM(q, Y_D)$<br><span style="margin-left: 150px;">(-)</span> <span style="margin-left: 100px;">(+)</span>                                                                  | Behavioral            |
| CA       | Current account           | $CA \equiv EX - IM = CA(q, Y_D, Y^*)$<br><span style="margin-left: 150px;">(+)</span> <span style="margin-left: 100px;">(-)</span> <span style="margin-left: 100px;">(+)</span> | Identity              |
| $D$      | Demand for domestic goods | $D \equiv C + I + G + CA$                                                                                                                                                       | Identity              |
| $A$      | Domestic demand           | $A \equiv C + I + G$                                                                                                                                                            | Identity              |
| $C$      | Consumption               | $C = C(Y_D)$<br><span style="margin-left: 150px;">(+)</span>                                                                                                                    | Behavioral            |
| $q$      | Real exchange rate        | $q \equiv \frac{EP^*}{P}$                                                                                                                                                       | Identity              |

# DD Curve

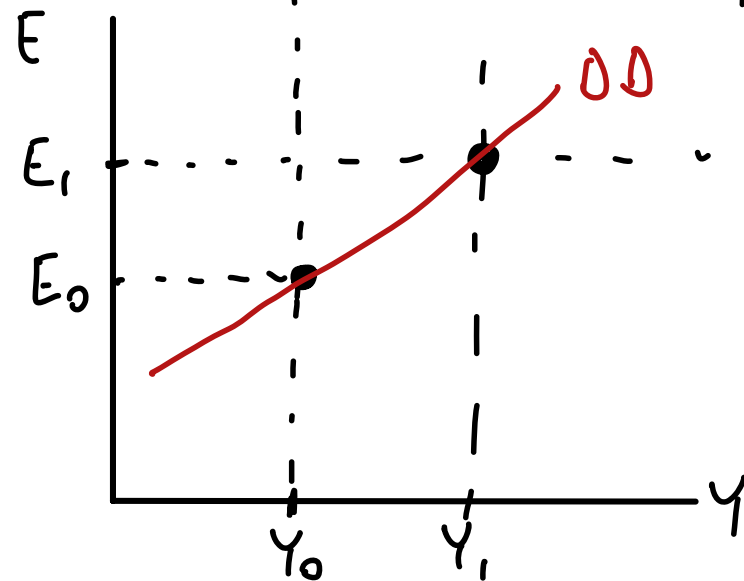
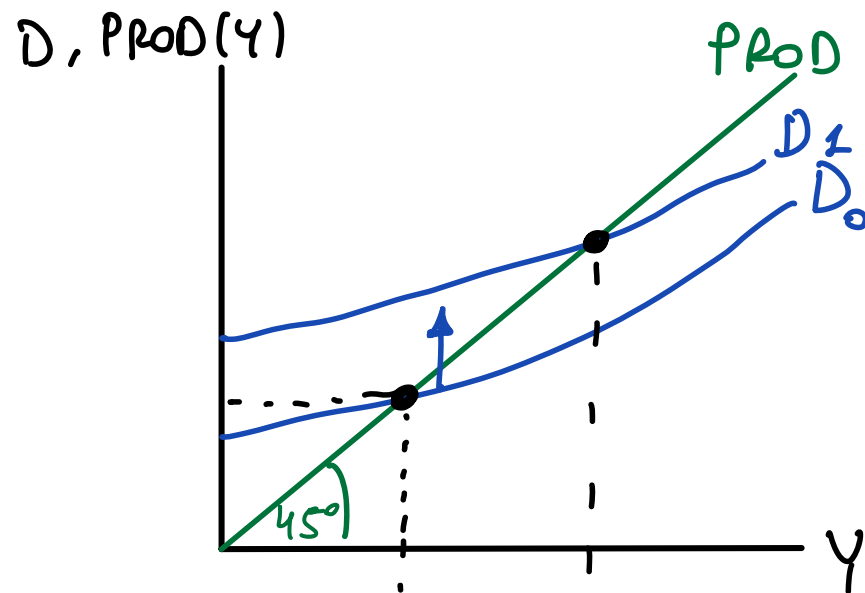
$$Y = D$$

$$Y = C + I + G + CA$$

$$Y = C_{(+)}(Y_D) + I + G + CA_{(+)}(q_{(+)} , Y_{D(-)} , Y^*_{(+)})$$

$$Y = C_{(+)}(Y_D) + I + G + CA_{(+)}(EP^*/P_{(+)} , Y_{D(-)} , Y^*_{(+)})$$

$Y \uparrow \quad C \uparrow$  BUT  $CA \downarrow$ . HOWEVER,  
 $D$  ALWAYS  $\uparrow$



$$D = C + I + G + NX \quad \text{WHEN } Y \uparrow$$

$C \uparrow$   $NX \downarrow$  SO  $D$ ?

NOT UNCLEAR,  $D \uparrow$

$C$ : CONS. OF DOM. AND FOREIGN GOODS

$C - IM$  = DOM DEM. FOR DOM GOOD

$$NX = \underbrace{EX}_{\text{DOES NOT CHANGE}} - \underbrace{IM}_{\text{IM GOES UP}}$$

DOES NOT  
CHANGE

IM GOES  
UP

$$C = C_{\text{DOM GOOD}} + C_{\text{FOREIGN GOOD}}$$

$IM$  = DOM DEMAND FOR  
FOREIGN

# Short-run FX and money market model

## Exogenous variables

| Variable | Description            |
|----------|------------------------|
| $R^*$    | Foreign interest rate  |
| $E^e$    | Expected exchange rate |
| $Y$      | Real income            |
| $M^s$    | Money supply           |
| $P$      | Price level            |

## Endogenous variables

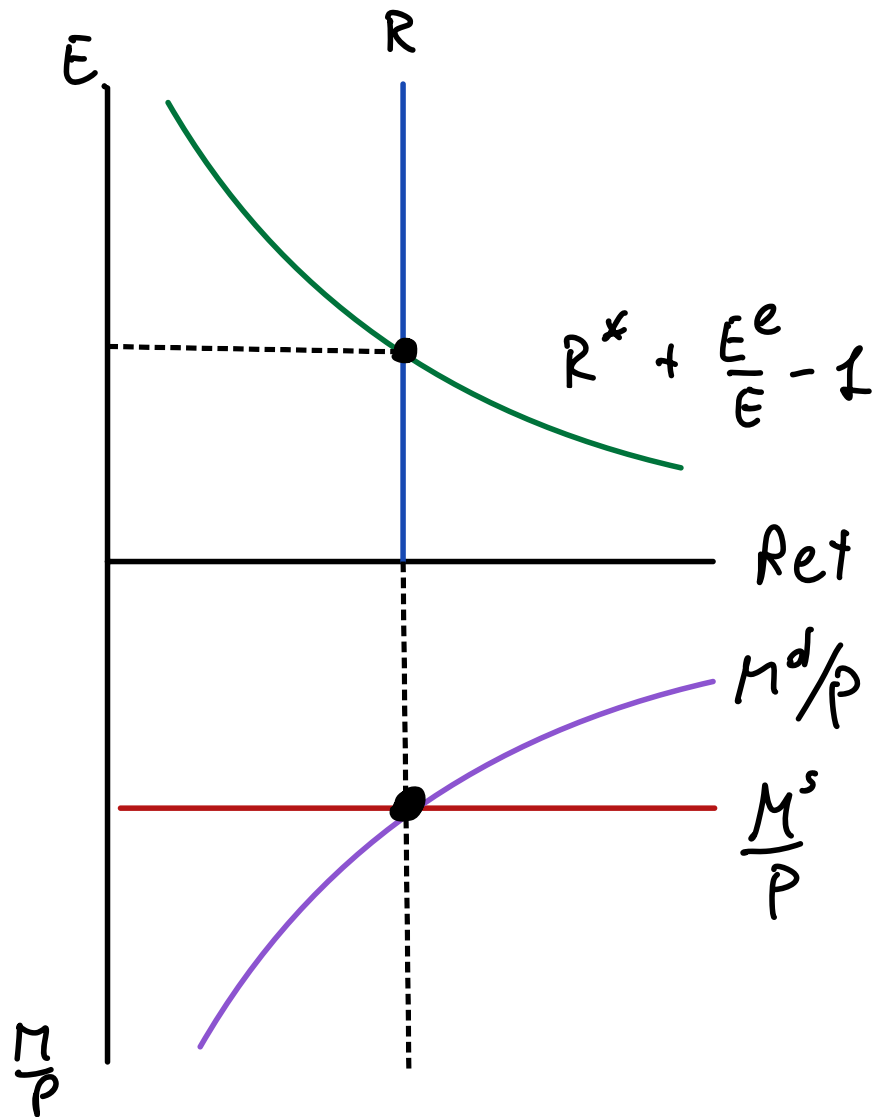
| Variable | Description            | Equation                      | Type of equation      |
|----------|------------------------|-------------------------------|-----------------------|
| $E$      | Exchange rate          | $R = R^* + \frac{E^e - E}{E}$ | Equilibrium condition |
| $R$      | Domestic interest rate | $M^d/P = L(R, Y)$             | Behavioral equation   |
| $M^d$    | Money demand           | $M^d = M^s$                   | Equilibrium condition |

# AA Curve

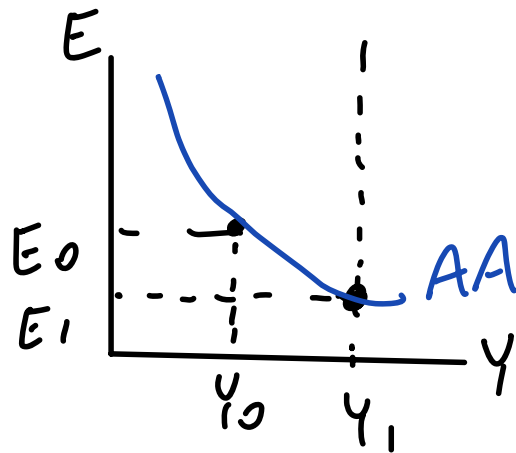
(UIP):  $R = R^* + \frac{E^e}{E} - 1$

(MS) = (MD):  $\frac{M^s}{P} = L(R, Y)$   
 (-) (+)

Y EXO } FIND E FOR  
 E ENDO } GIVEN Y



# AA Curve

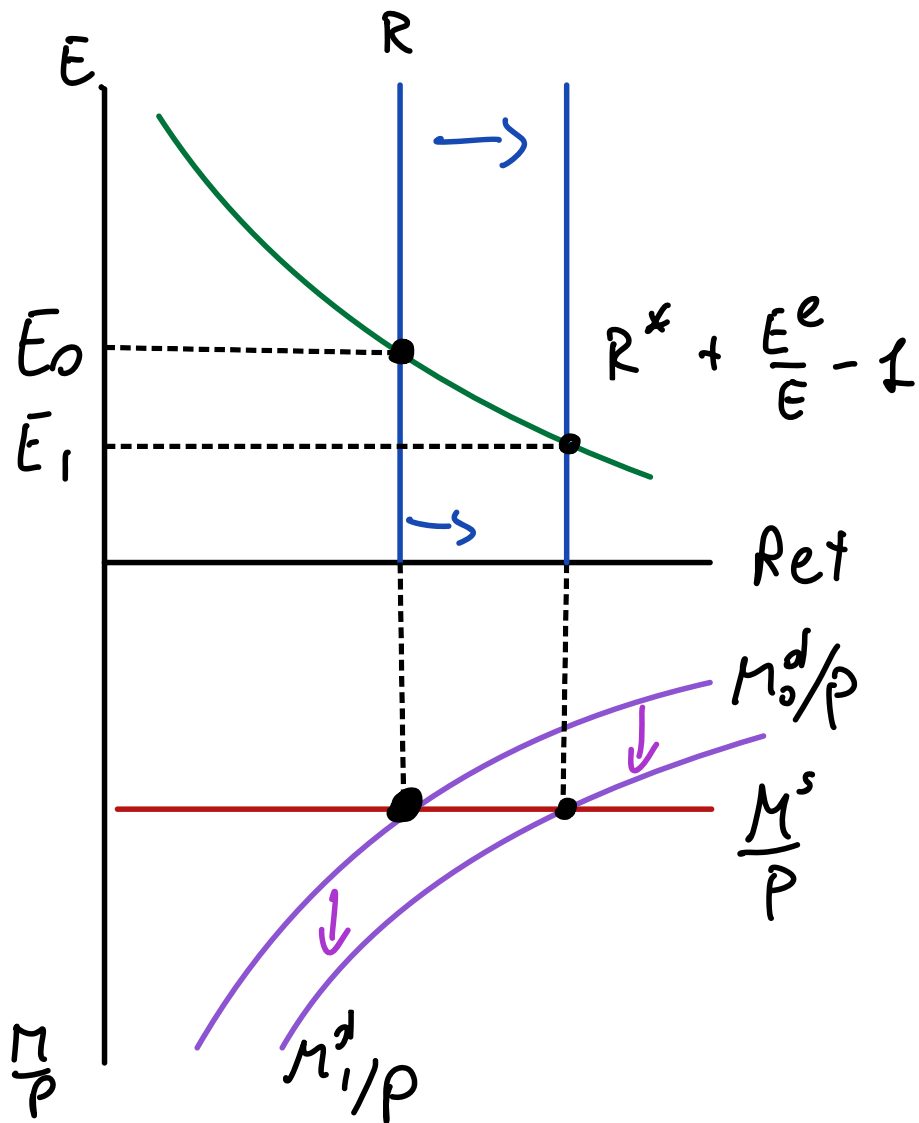


(UIP): 
$$R = R^* + \frac{E^e}{E} - 1$$

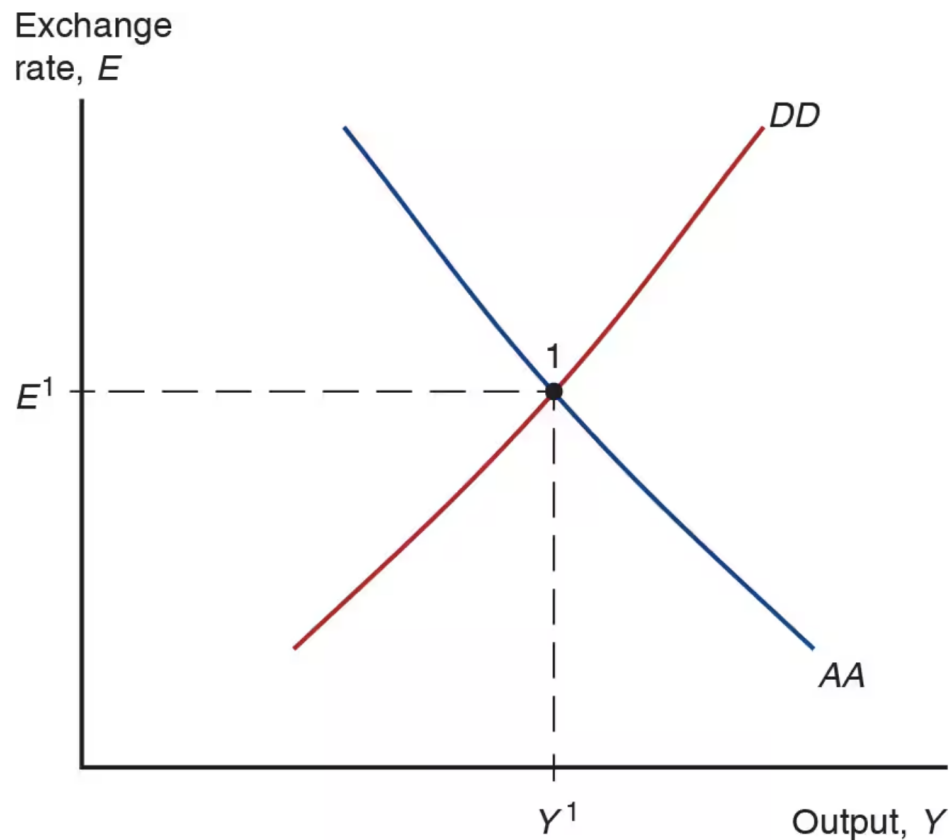
(MS) = (MD): 
$$\frac{M^s}{P} = L(R, Y)$$
  
 (-) (+)

$(Y_0, E_0)$  WITH  $Y_1 > Y_0$

$(Y_1, E_1)$   $E_1 < E_0$



# Short-run Equilibrium in *AA-DD* model



GOODS

$E$  EXO

MM + FX

$Y$  EXO

COMBINE

$Y, E$  ARE

BOTH

ENDO

# Example: DD Schedule

$$C_{(+)}(Y_D) = 1 + 0.75Y_D$$

$$EX_{(+)}(q, Y_{(+)}^*) = 0.15 + 0.5q + 0.1Y^*$$

$$IM_{(-)}(q, Y_{(+)}^D) = 0.1 - 0.2q + 0.12Y_D$$

$$Y_D = Y - T$$

$$q = \frac{EP^*}{P}$$

How to find the DD schedule

EQ. COND FOR GOODS:

$$Y = D$$

$$Y = C + I + G + CA$$

$$Y = 1 + 0.75(Y - T) + I + G + 0.15 + 0.5q + 0.1Y^* - 0.1 + 0.2q - 0.12(Y - T)$$

# Example: DD Schedule

. CAN SOLVE FOR E  
TO GET EXPLICIT  
E = FUNCTION OF Y.

$$C(Y_D) = 1 + 0.75Y_D$$

(+)

$$EX(q, Y^*) = 0.15 + 0.5q + 0.1Y^*$$

(+)

$$IM(q, Y_D) = 0.1 - 0.2q + 0.12Y_D$$

(-) (+)

$$Y_D = Y - T$$

$$q = \frac{EP^*}{P}$$

How to find the DD schedule

$$Y = 1 + 0.75(Y - T) + I + G +$$

$$+ 0.15 + 0.5q + 0.1Y^*$$

$$- 0.1 + 0.2q - 0.12(Y - T)$$

$$Y = 1 + 0.15 - 0.1 + I + G$$

$$(0.75 - 0.12)(Y - T) +$$

$$(0.5 + 0.2) \frac{EP^*}{P} + 0.1Y^*$$

# Example: AA Schedule

• CAN SOLVE FOR  $E$   
TO GET:

$E = \text{FUNCTION OF } Y$

$$R = R^* + \frac{E^e - E}{E}$$

$$\frac{M^d}{P} = \frac{M^s}{P}$$

$$\begin{aligned} \frac{M^d}{P} &= L\left(\underset{(-)}{R}, \underset{(+)}{Y}\right) \\ &= 1.1 - R + 0.05Y \end{aligned}$$

How to find the AA schedule

$$\frac{M^s}{P} = \frac{M^d}{P}$$

$$\frac{M^s}{P} = 1.1 - R + 0.05Y$$

$$\frac{M^s}{P} = 1.1 - R^* - \frac{E^e}{E} + 1 + 0.05Y$$

# Example: AA Schedule

• CAN SOLVE FOR  $E$   
TO GET:

$E = \text{FUNCTION OF } Y$

$$R = R^* + \frac{E^e - E}{E}$$

How to find the AA schedule

$$\frac{M^d}{P} = \frac{M^s}{P}$$

$$\frac{M^s}{P} = 1.1 - R^* - \frac{E^e}{E} + 1 + 0.5Y$$

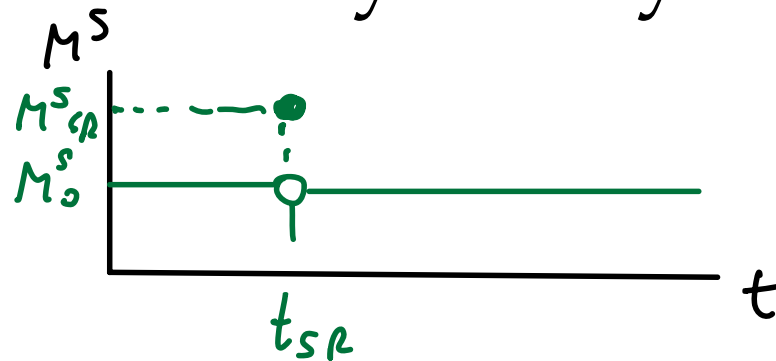
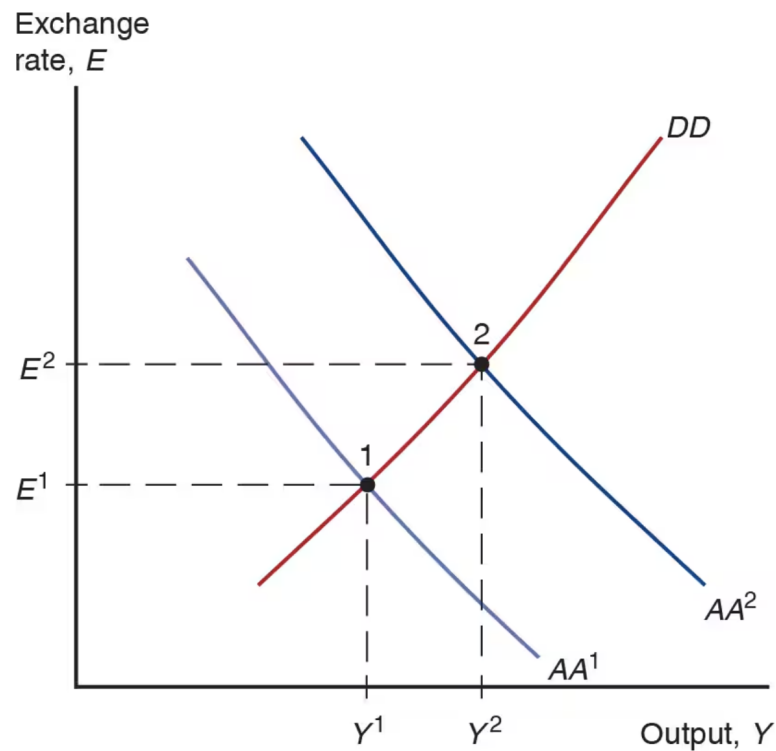
$$\begin{aligned} \frac{M^d}{P} &= L(\underset{(-)}{R}, \underset{(+)}{Y}) \\ &= 1.1 - R + 0.05Y \end{aligned}$$

SOLVE FOR  $E$ :

$$E^e/E = 1.1 - R^* + 1 - M^s/P + 0.5Y$$

$$E = \frac{E^e}{1.1 - R^* + 1 - M^s/P + 0.5Y}$$

# Temporary Change in Monetary Policy



TEMPORARY SHOCKS

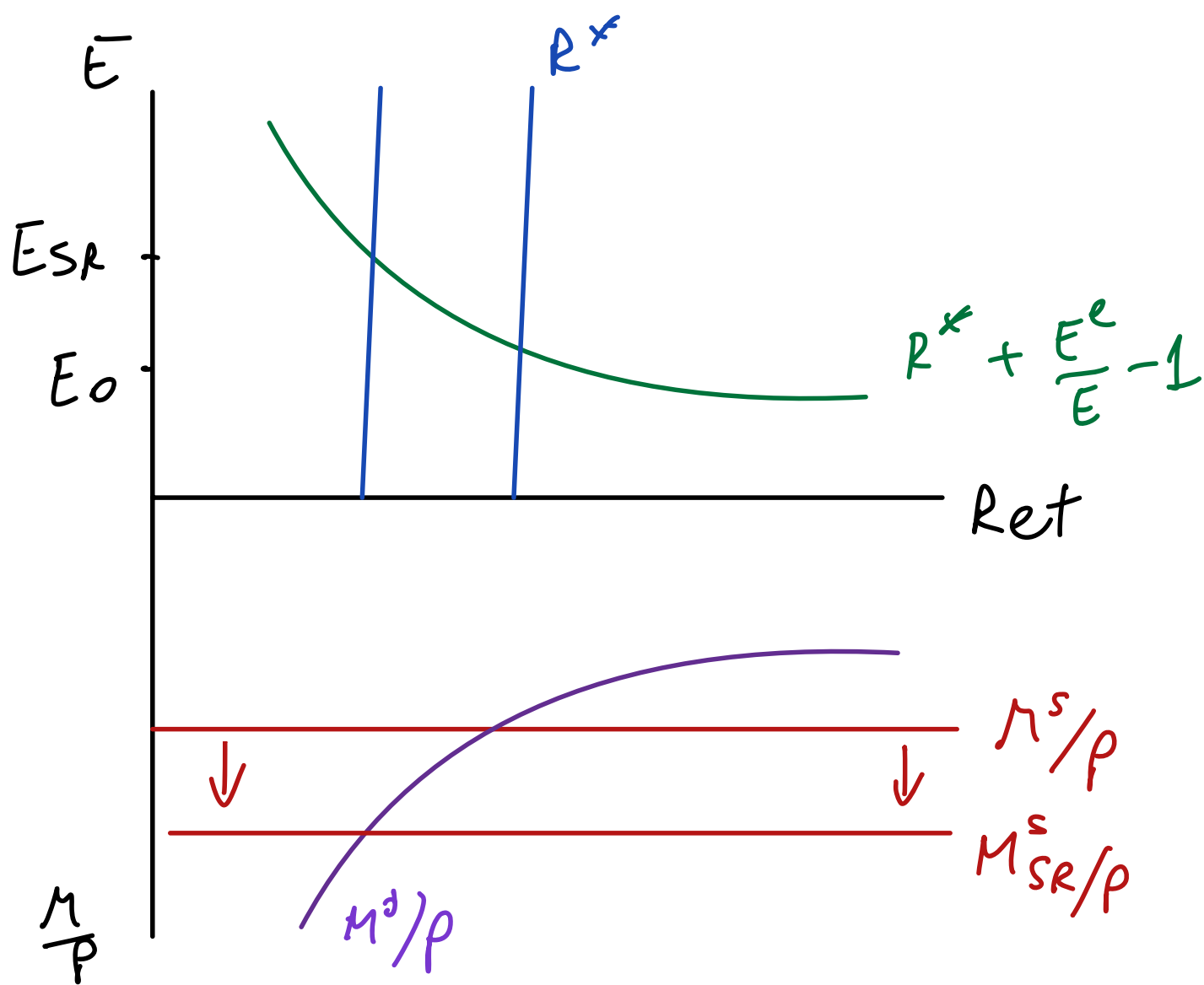
- KEEP LONG RUN  $E$

UNCHANGED  $\Rightarrow$

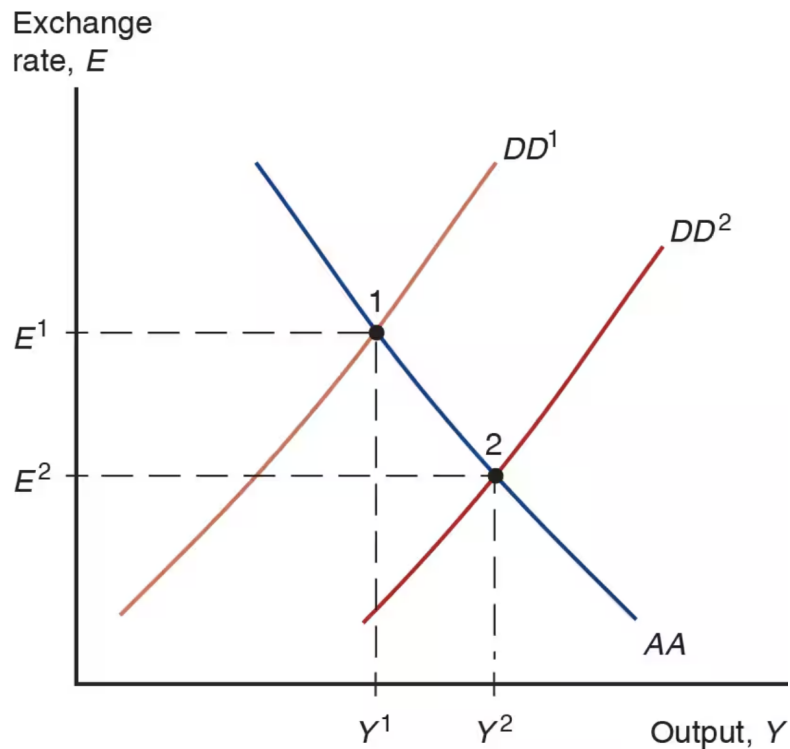
$E_{SR}^e = E_{LR}$  UNCHANGED

- $P_{SR} = P_0$  FIXED

$M^S \uparrow$   
 LEADS TO  
 $E \uparrow$



# Temporary Change in Fiscal Policy



$G \uparrow$  IN SR ONLY

$D \uparrow$   $Y \uparrow$  FOR ANY  $E$

$\Rightarrow$  DD SHIFTS TO  
THE RIGHT

# Example: DD Schedule

$$C_{(+)}(Y_D) = 1 + 0.75Y_D$$

$$EX_{(+)}(q, Y^*)_{(+)} = 0.15 + 0.5q + 0.1Y^*$$

$$IM_{(-)}(q, Y_D)_{(+)} = 0.1 - 0.2q + 0.12Y_D$$

$$Y_D = Y - T$$

$$q = \frac{EP^*}{P}$$

- Plug in given  $C$ ,  $EX$ ,  $IM$ ,  $Y_D$ ,  $q$  into the equilibrium condition for the goods market  $Y = C + I + G + EX - IM$  to get:

$$Y = 1.05 + 0.63(Y - T) + 0.7 \frac{EP^*}{P} + I + G + 0.1Y^*$$

- Solve for  $E$  to get the DD curve:

$$E = \frac{1}{0.7} \frac{P}{P^*} (-1.05 + 0.37Y + 0.63T - I - G - 0.1Y^*)$$

# Example: AA Schedule

$$R = R^* + \frac{E^e}{E} - 1$$

$$\frac{M^d}{P} = \frac{M^s}{P}$$

$$\frac{M^d}{P} = L(\underset{(-)}{R}, \underset{(+)}{Y}) = 1.1 - R + 0.05Y$$

- Plug in UIP into money market equilibrium condition:

$$\frac{M^s}{P} = 1.1 - \left( R^* + \frac{E^e}{E} - 1 \right) + 0.05Y$$

- Solve for  $E$  to get the AA curve:

$$E = \frac{E^e}{2.1 - \frac{M^s}{P} - R^* + 0.05Y}$$

# Full AA-DD Model

DD Schedule:  $Y = C(Y - T) + I + G + CA(EP^*/P, Y - T, Y^*)$

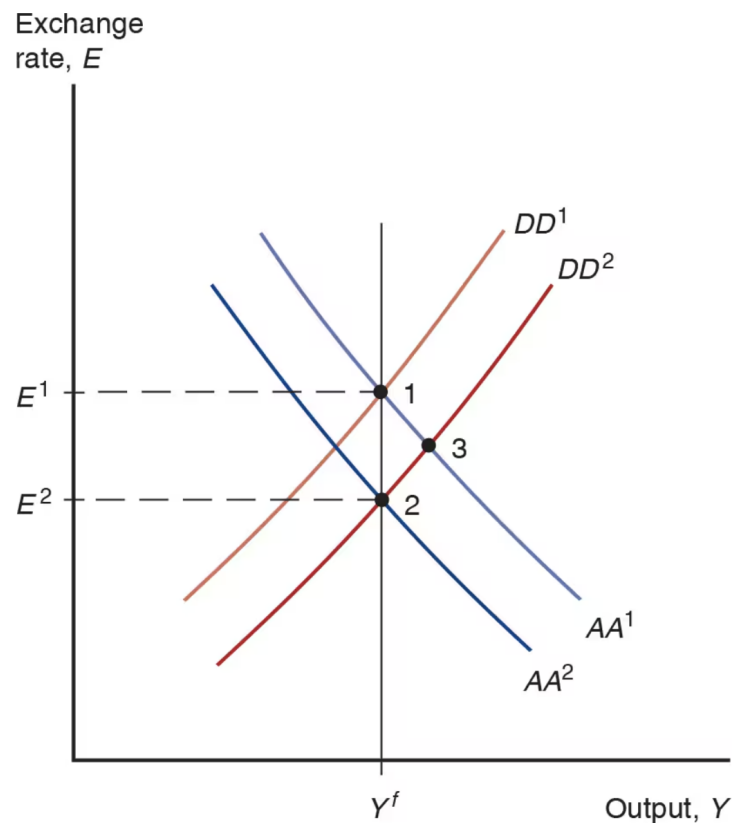
AA Schedule:  $\frac{M^s}{P} = L\left(R^* + \frac{E^e}{E} - 1, Y\right)$

Phillips Curve:  $\pi = \pi^e + \alpha(Y - Y^f)$

Definition of inflation:  $\pi_t = \frac{P_t}{P_{t-1}} - 1$

Definition of expected inflation:  $\pi_t^e = \frac{P_{t+1}^e}{P_t} - 1$

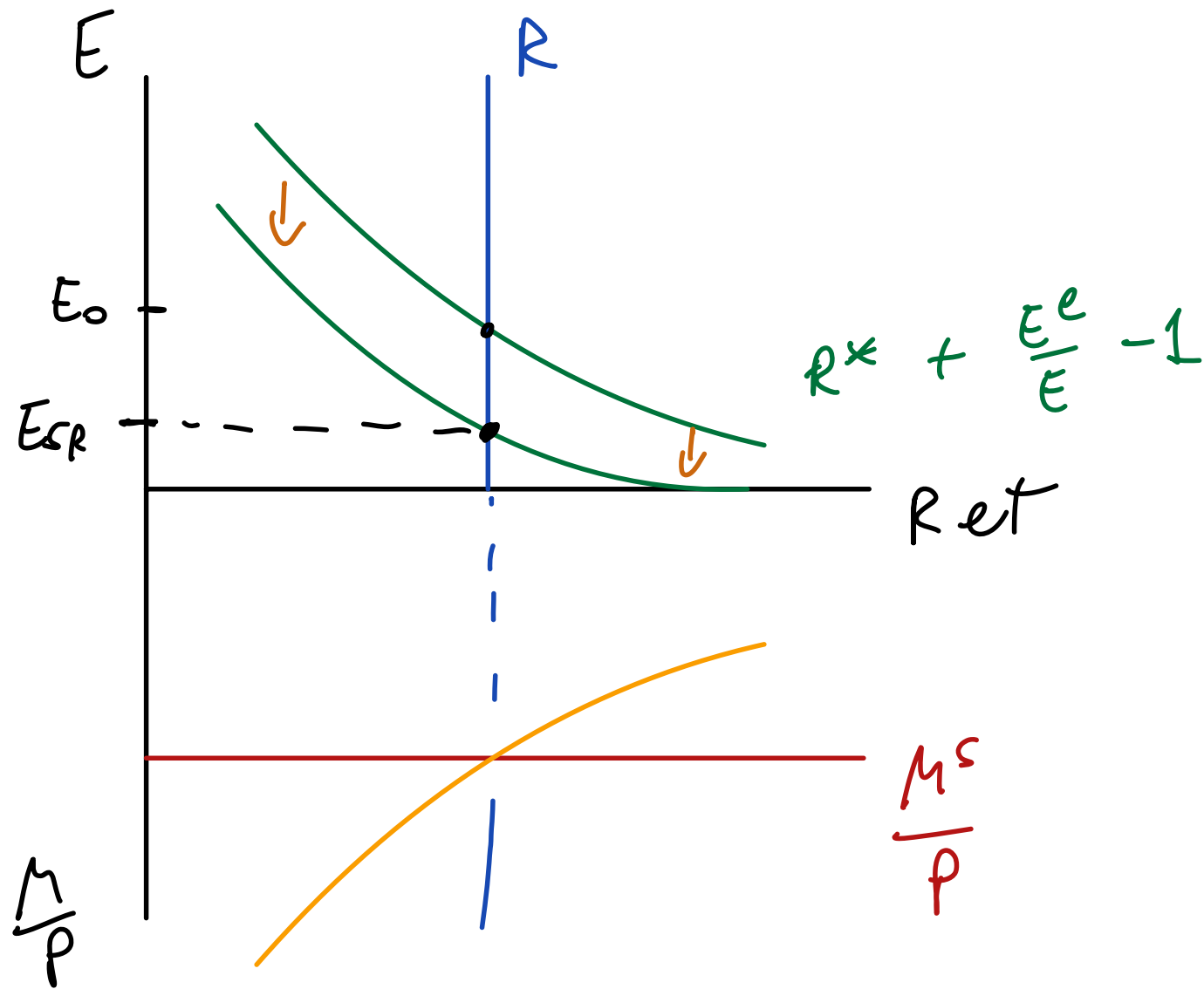
# Permanent Shifts in Fiscal Policy



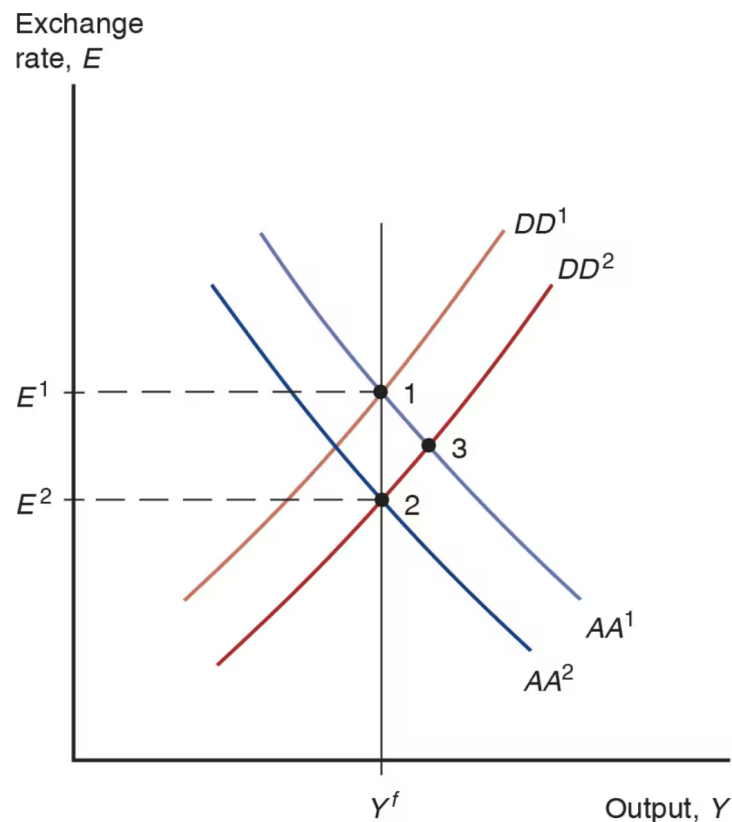
$G \uparrow$  PERMANENTLY

$\Rightarrow D \uparrow \Rightarrow Y \uparrow \Rightarrow DD$  SHIFTS  
TO THE  
RIGHT

$$(AA) : \frac{M^s}{P} = L \left( R^* + \frac{E^e - E}{E}, Y \right)$$

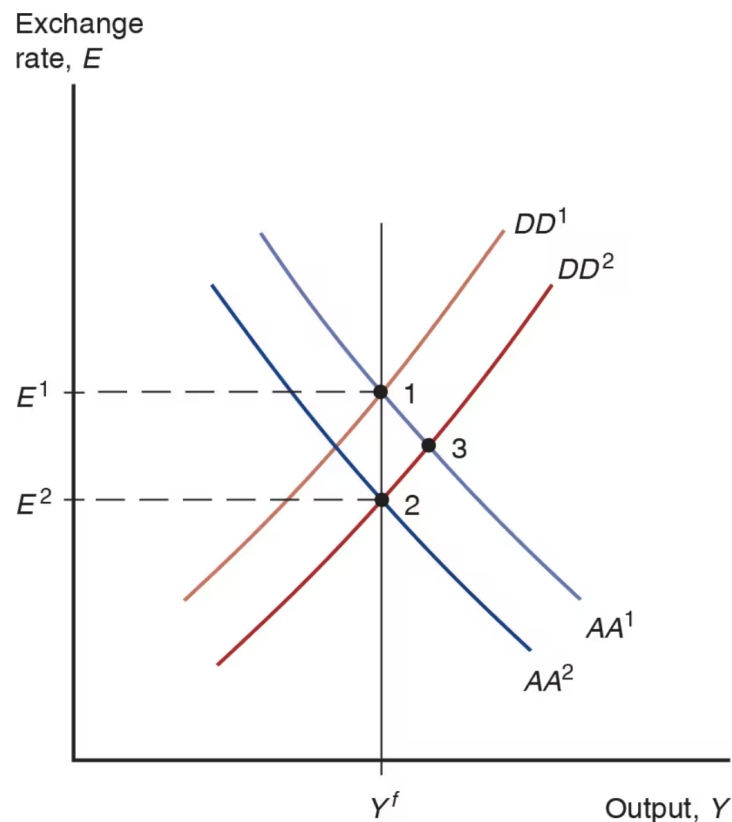


# Permanent Shifts in Fiscal Policy



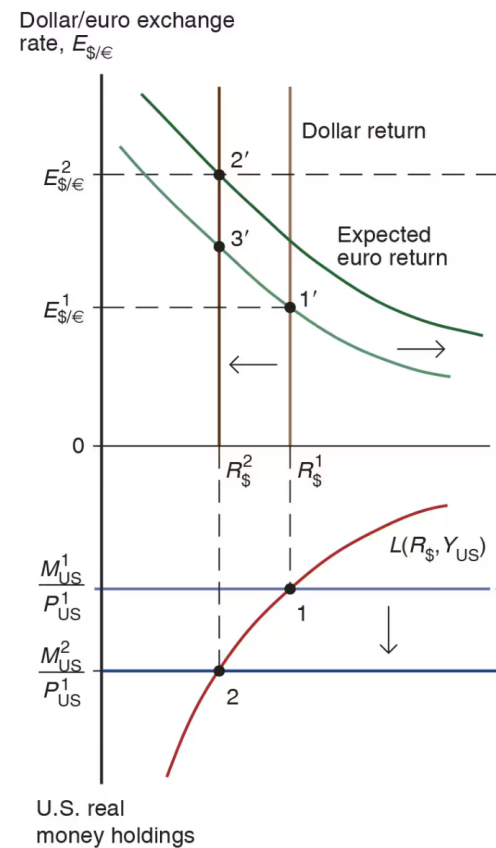
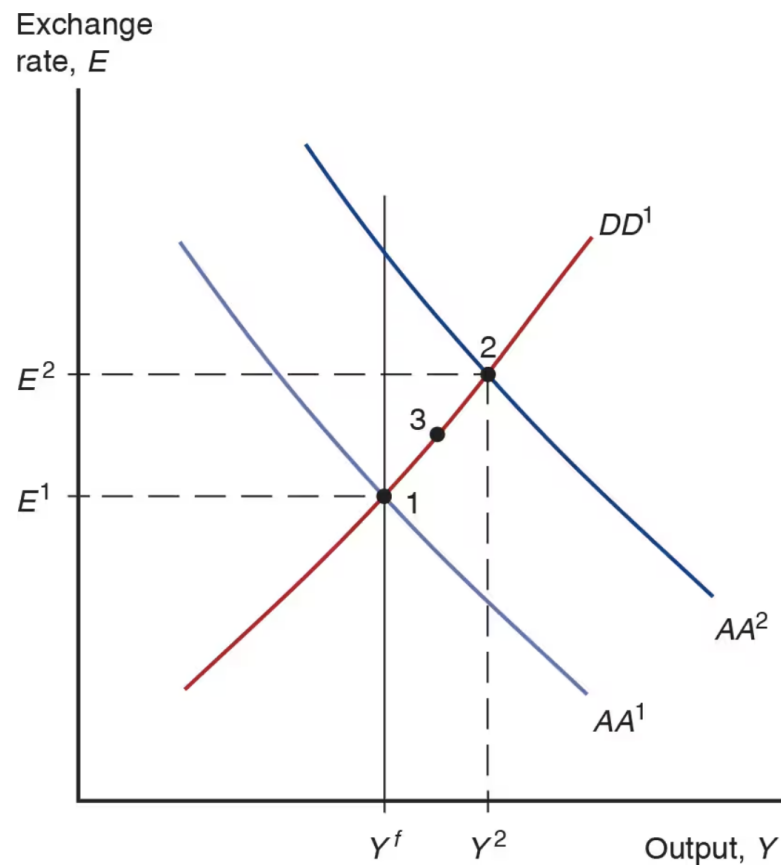
- Start in a medium run equilibrium at point 1 with  $R_0 = R^*$  and  $E_0^e = E^1$
- Increase in  $G$  shifts  $DD$  to the right
- At point 3, exchange rate is lower than  $E_1$
- Since the shift in  $DD$  is permanent, appreciation at point 3 is also expected to be permanent
- $E^e$  must also go down
- Lower  $E^e$  shifts  $AA$  down
- But how much?

# Permanent Shifts in Fiscal Policy



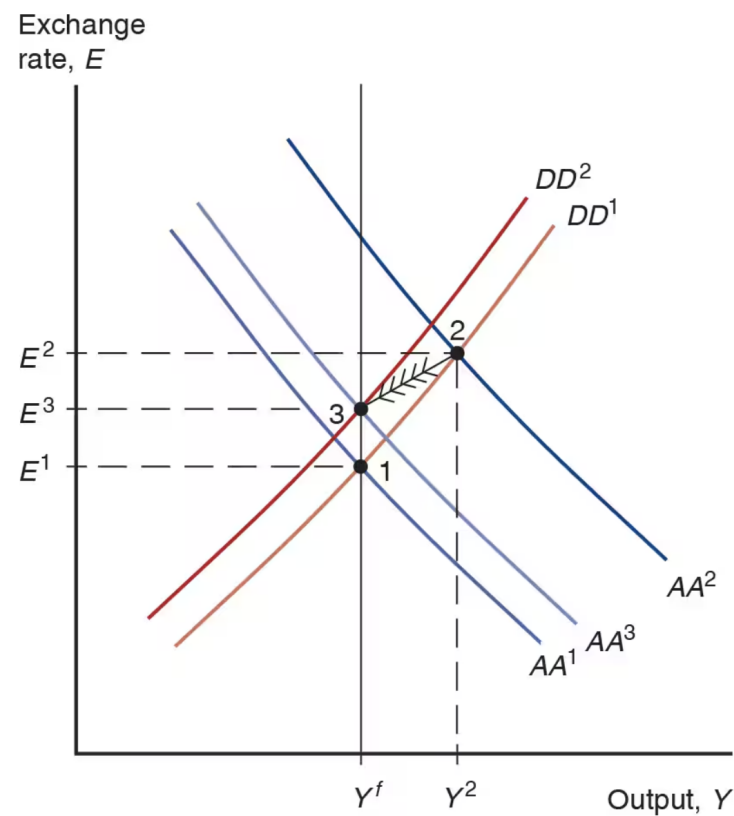
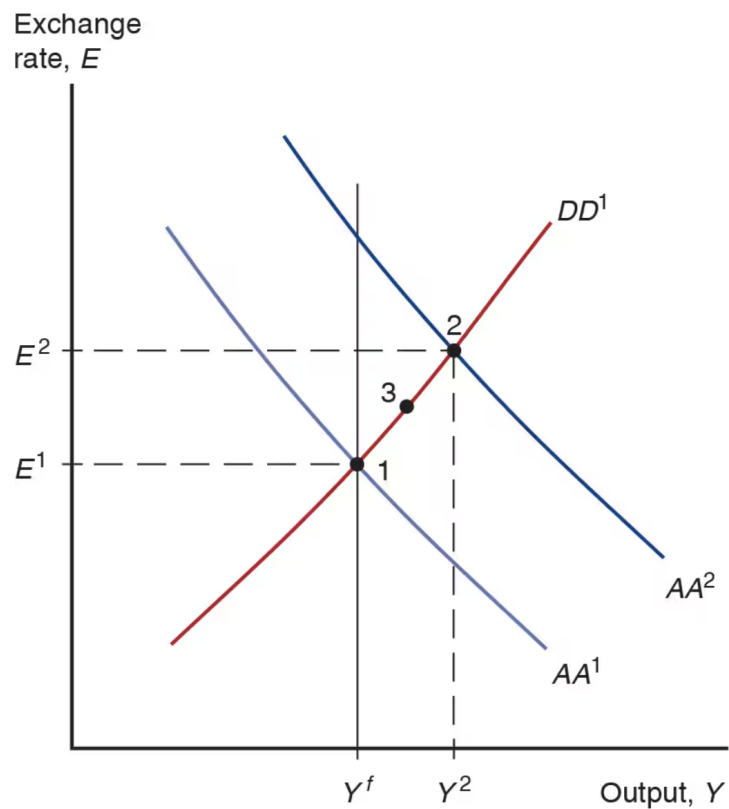
- There are no more shifts expected to occur,  $E^e$  remains constant
- Real money supply is constant
- Long run real money demand must stay unchanged
- Then  $P$  never changes
- $MS=MD$  implies  $R$  does not change
- UIP implies  $E$  does not change either
- If  $Y$  must be equal to its original level  $Y^f$ ,  $AA^1$  must shift to  $AA^2$

# Permanent Shifts in Monetary Policy

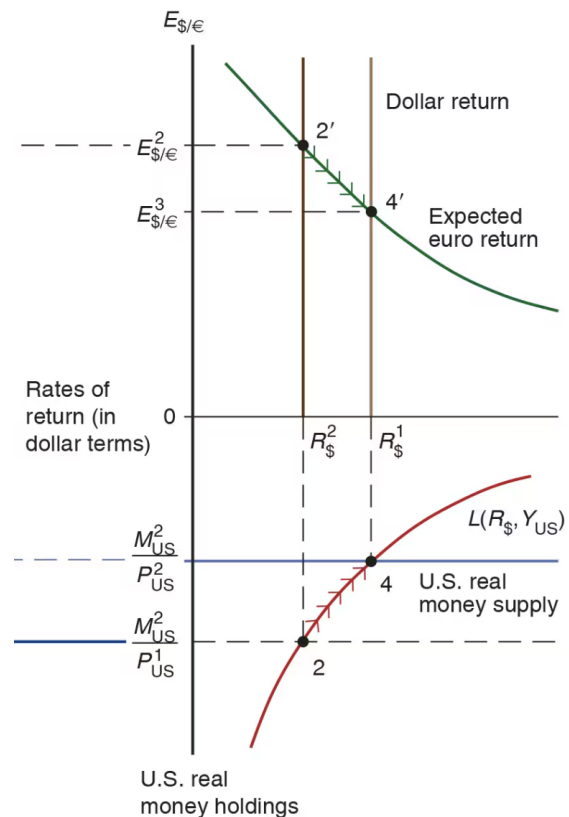
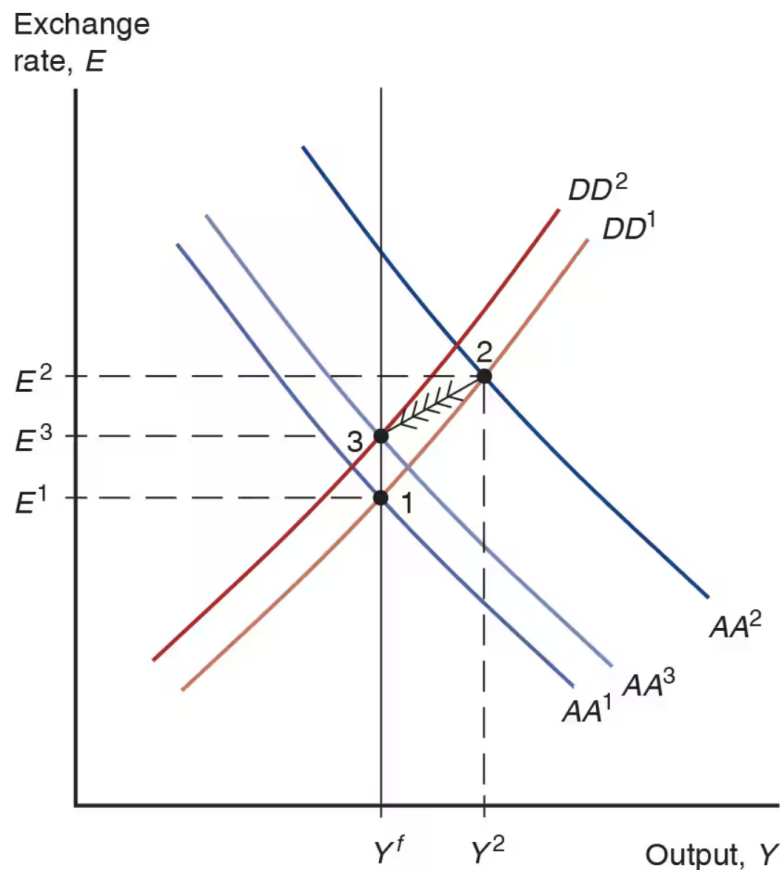


(a) Short-run effects

# Permanent Shifts in Monetary Policy



# Permanent Shifts in Monetary Policy



(b) Adjustment to long-run equilibrium

# DD Schedule with Tariffs

IMPORT TARIFFS - "AD VALOREM"  $\rightarrow$  PROPORTIONAL

$$\begin{array}{l|l} \underbrace{IM}_{\substack{\text{UNITS} \\ \text{OF DOM} \\ \text{GOOD}}} = q \times \underbrace{VOLUME}_{\substack{\text{IN UNITS} \\ \text{OF FOREIGN} \\ \text{GOOD}}} & IM(q, Y-T) \\ & \begin{array}{c} (-) \quad (+) \\ = q \times VOLUME(q, Y-T) \\ \quad \quad \quad (-) \quad (+) \\ = q \times VOLUME\left(\frac{EP^*}{P}, Y-T\right) \end{array} \end{array}$$

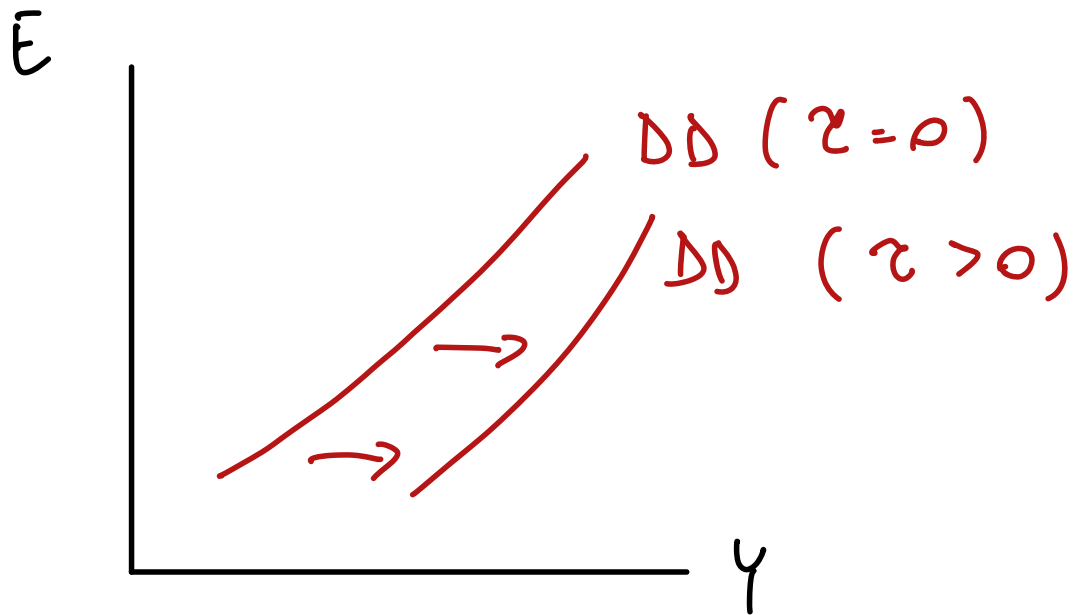
$\downarrow$   
CHANGE OF UNITS

$$IM = q \times VOLUME \left( \frac{EP^x}{P} (1 + \tau) , Y - T \right)$$

AD VALOREM TARIFF

$$IM = IM \left( \underset{(-)}{q} , \underset{(+)}{Y - T} , \underset{(-)}{\tau} \right)$$

$$Y = C + I + G + EX - IM \left( \underset{(-)}{\tau} \right)$$



DD SHIFTS TO  
THE RIGHT  
WHEN  $z$   
GOES UP